

PretrainZero: Reinforcement Active Learning on Pretraining Data

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Abstract

Recent reinforcement learning (RL) based large-thinking models demonstrate impressive expert-level abilities, i.e., software and math, but still rely heavily on verifiable rewards in specific domains, which places a significant bottleneck to extend the performance boundary of general reasoning capabilities. In this work, we propose PretrainZero, a reinforcement active learning framework built on the pretraining corpus to extend RL from domain-specific post-training to general pretraining. PretrainZero features the following characteristics:

1) **Active pretraining objectives:** inspired by the active learning ability of humans, PretrainZero learns a unified reasoning policy to actively identify reasonable and informative contents from pretraining corpus, and reason to predict these contents by RL. 2) **Self-supervised learning:** without any verifiable labels, pretrained reward models, or supervised fine-tuning, we directly pre-train reasoners from 3 ~ 30B base models on the general Wikipedia corpus using RL, significantly breaking the verification data-wall for general reasoning. 3) **Verification scaling:** by tackling increasingly challenging masked spans, PretrainZero substantially enhances the general reasoning abilities of pretrained base models. With reinforcement learning on pretraining data, PretrainZero improves Qwen3-4B-Base for 8.43, 5.96 and 10.60 on MMLU-Pro, SuperGPQA and math average benchmarks. In post-training, the pretrained models can also serve as reasoning foundation models for downstream RLVR tasks.

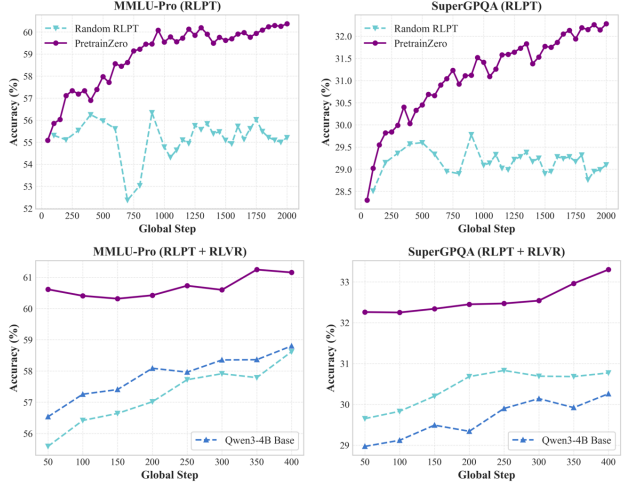


Figure 1. Performance of Reinforcement Pre-Training on pretraining data (RLPT) and with post-training (RLPT + RLVR) stages.

1. Introduction

Recent large language models (LLMs) have achieved human-level expertise in specific domains, particularly through large-scale self-supervised learning in pretraining (Kaplan et al., 2020; Achiam et al., 2023) and Reinforcement Learning (RL) (Guo et al., 2025; Yu et al., 2025a; Chu et al., 2025) in post-training. During pretraining, self-supervised learning with a fixed next-token prediction paradigm allows models to leverage large-scale, low-cost data to improve general capabilities effectively. In contrast, the post-training RL faces a severe data-wall: Reinforcement Learning with Verifiable Rewards (RLVR) (Guo et al., 2025; Yue et al.) requires domain-specific verifiers to label training samples, and Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022), relying on reward models and humans, can only train limited steps to avoid reward hacking. This motivates a natural direction—performing reinforcement learning (Dong et al., 2025; Li et al., 2025) in a self-supervised pretraining manner (Brown et al., 2020), in order to use inexpensive pretraining data to extend RLVR and overcome this data-wall.

However, formulating the self-supervised pretraining as RLVR tasks is non-trivial. This work first investigates stand-alone Reinforcement Learning on Pre-Training data (RLPT) (Dong et al., 2025) under three principles: 1) Comprehensiveness: we establish both baselines including masked

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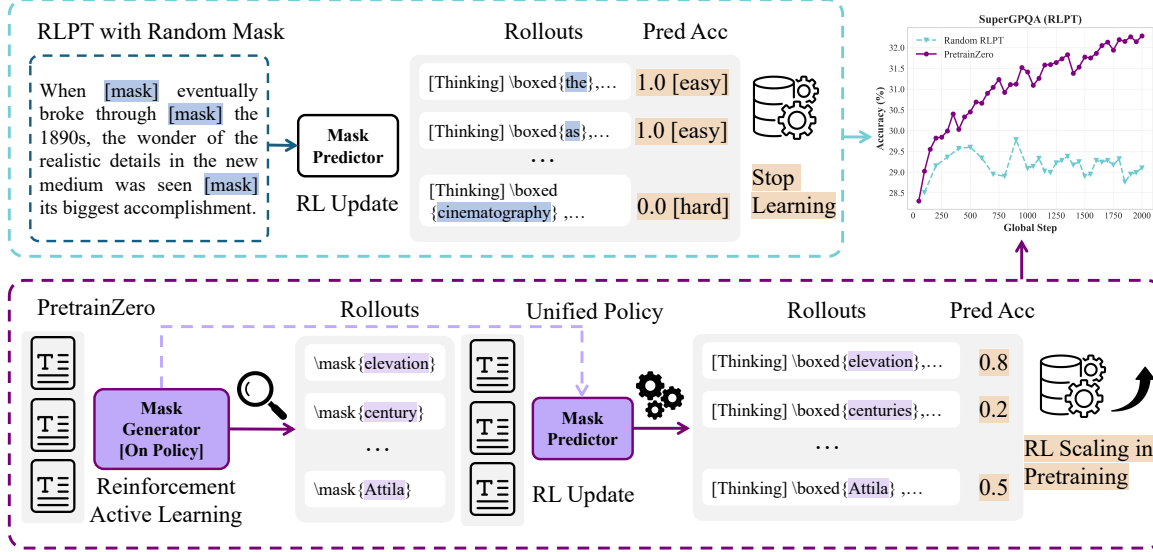


Figure 2. An overview of reinforcement active learning on Pretraining data. Compared with vanilla RLPT, PretrainZero actively explores and learns from the informative contexts on the pretraining corpus.

token prediction and next token prediction as the reasoning objective (Dong et al., 2025). 2) Full self-supervision: we exclude any additional Supervised Fine-Tuning (SFT) dataset and reward models (Li et al., 2025). 3) Generalization: we avoid Question–Answer (QA) formats or synthetic chain-of-thought (CoT) datasets, like OmniMath (Gao et al., 2024), and train directly on general-domain Wikipedia (Devlin et al., 2019). Experimental results demonstrate that the vanilla RLPT fails to elicit high-quality CoT: the low information density of pretraining corpus leads to ineffective learning, and the noisy and incorrect tokens often causes training collapse.

This work proposes the first stand-alone RLPT method to extend RLVR on real-world pretraining corpus, termed PretrainZero. This is achieved by mimicking the human active-learning behavior (Yang et al., 2025b; Settles, 2009): humans can actively learn from a broad range of experiences, selectively focusing on informative elements and unfamiliar concepts. This allows effective learning even when the underlying experiences are noisy or low in information density. In contrast, current large language models, whether through supervised learning or RLPT, rely on fixed prediction patterns, such as next-token or off-policy selected masked-token prediction. These rigid learning patterns limit their efficiency (Berglund et al.) and prevent them from leveraging data as flexibly as humans do.

Inspired by this, this work proposes a reinforcement active learning framework in order to learn from real-world pretraining distributions. Specifically, we introduce an on-policy *mask generation task* as an auxiliary active-learning objective for the *masked-span prediction task*. The mask generation policy actively explores informative, verifiable,

and not-yet-mastered content within the pretraining data and selects it as learning targets. Consequently, the masked-span prediction policy learns to produce CoT reasoning and recover these informative spans. For optimization, we formulate a min–max bilevel reinforcement learning objective, where each batch is jointly optimized using GRPO over both the mask generation and masked-span prediction tasks. Different from unsupervised RL methods such as self-play (Huang et al., 2025a) and test-time scaling (Snell et al., 2024), PretrainZero offers a **verifiable RL scaling mechanism grounded in real data** via self-supervision. This avoids the hallucination and training collapse caused by relying on majority voting over model-generated answers in prolonged RL training (Liu et al., 2025).

As shown in Figure 1, we evaluate RLPT for 2000 steps in pretraining and add general RLVR in post-training. For a Qwen3-4B-Base (Yang et al., 2025a) model, PretrainZero consistently improves 8.43, 5.96, and 10.60 on MMLU-Pro (Wang et al., 2024a), SuperGPQA (Du et al., 2025), and math average benchmarks during RLPT. After general RLVR (Ma et al., 2025) in the post-training stage, these improvements remain substantial, with final improvements of 2.35, 3.04, and 2.81 on MMLU-Pro, SuperGPQA, and math average respectively.

Our contributions are summarized as follows:

- We introduce PretrainZero, the first stand-alone RLPT method to operate RLVR on real-world pretraining corpus, enabling general-domain and large-scale RL directly from base models using only pretraining data as grounding.
- We propose the reinforcement active learning mechanism inspired by human active learning. The mask-generation objective enables the model to actively identify what in-

formation to learn, improving training efficiency in low-information-density pretraining corpora.

- We evaluate PretrainZero in both the continue-pretraining and post-training stages, showing that it mitigates the general reasoning data-wall with pretraining data, and finally the pretrained reasoning models can serve as the reasoning foundation models for general downstream RLVR tasks.

2. Preliminary

In order to learn from the pretraining corpus, traditional self-supervised pretraining adopts language modeling objectives to capture linguistic patterns and contextual dependencies. Recently, the emerging reinforcement learning on pretraining data (RLPT) constructs verifiable data through token prediction to learn the reasoning process within concepts. We briefly review different learning patterns in this section.

2.1. Self-Supervised Pretraining

Given the context, traditional self-supervised pretraining tasks include masked token prediction (MTP) (Radford et al., 2019) and next token prediction (NTP) (Devlin et al., 2019). As shown in Eq. 1, the NTP task predict the identity tokens x_t in each location given their preceding context $x_{<t}$ under an auto-regressive pattern:

$$\mathcal{J}_{\text{NTP}}(\theta) = \sum_{t=1}^T \log \pi_{\theta}(x_t | x_{<t}), \quad (1)$$

where x is the token sequence with length T and θ is the pretrained model parameters. As the counterpart, masked token prediction task jointly leverages both the preceding and succeeding contexts $x_{m<t,t>n}$ to predict the masked tokens $x_{m \leq t \leq n}$:

$$\mathcal{J}_{\text{MTP}}(\theta) = \sum_{t=m}^n \log \pi_{\theta}(x_{m \leq t \leq n} | x_{m < t, t > n}). \quad (2)$$

Supported by self-supervised pretraining, modern LLMs (OpenAI, 2024; Liu et al., 2024) successfully scale up pretraining on massive Internet data. In this work, we simulate both masked token prediction and next token prediction as reinforcement reasoning tasks to explore more general RL approach (Ma et al., 2025).

2.2. Reinforcement Learning on Pretraining Data

Recent Reinforcement Pre-Training (RPT) (Dong et al., 2025) extends reinforcement learning into the pretraining corpus, constructing verifiable training data directly from the pretraining corpus and thereby alleviating the reliance on costly annotations and specific environments for verification. Specifically, RPT introduces the next-token reasoning task:

given a sequence x , one token x_t is treated as ground-truth and its preceding tokens $x_{<t}$ as context for the generated output, o_t . Unlike the self-supervised NTP task, where the model directly predicts the next token, RPT (Dong et al., 2025) first produces a chain-of-thought reasoning process (Guo et al., 2025) before generating the final predicted token. In optimization, RPT applies GRPO algorithm with group size G , and uses the exact match verifiable reward r_t^i between prediction $\hat{x}_t^i$ and ground-truth x_t^i :

$$\mathcal{J}_{\text{RPT}}(\theta) = \mathbb{E}_{(x_{\leq t}) \sim \mathcal{D}, \{o_t^i\}_{i=1}^G \sim \pi_{\theta}(\cdot | x_{<t})} [\mathbb{I}[\hat{x}_t^i = x_t^i]]. \quad (3)$$

Discussion on the weakness of vanilla RLPT. Despite the simplicity of RPT and its potential to extend the RLVR-style method into pretraining, several significant concerns also emerge, making vanilla RPT unsuitable for practical pretraining settings:

- **Robustness on real-world corpus:** Although RPT demonstrates improvements on synthetic dataset OmniMath, real-world pretraining data with more noise (Devlin et al., 2019) often causes training collapse.
- **Learning effectiveness:** Due to the low information density of pretraining corpora, simple token selection methods fails to identify informative content, leading to ineffective optimization.
- **Training efficiency:** Unlike self-supervised Next Token Prediction (NTP), which predicts all tokens in parallel, RPT predicts only a single token per sample, resulting in limited learning signal per training step.

3. PretrainZero

To solve these questions, this work first establishes a Reinforcement Learning on Pre-Training data (RLPT) baseline on the widely used general domain Wikipedia dataset with Qwen3-4B-Base model. Based on the empirical observations, we then propose a unified and active pretraining task to confirm the general and practical RLPT.

3.1. RLPT Baselines

We establish an RLPT baseline with three masking strategies for training corpus. The model is required to predict masked tokens x_t through CoT reasoning, receiving binary rewards from exact match with ground truth: $\mathcal{J}_{\text{RLPT}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, t, \{o_t^i\}_{i=1}^G \sim \pi_{\theta}(\cdot | x_{\setminus t})} [\mathbb{I}[\hat{x}_t^i = x_t^i]]$. Three mask prediction strategies for RLPT are investigated:

- **Random Next Token Reasoning.** The sequence is randomly truncated and the last token before truncation is masked for prediction. For each sample, the model first generates a CoT and only one selected token is predicted according to the generated CoT.

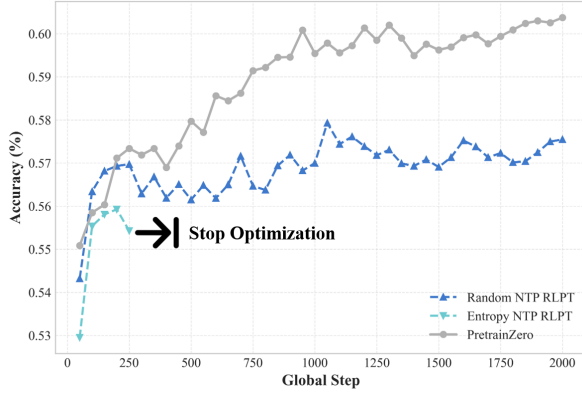


Figure 3. MMLU-Pro performance for Reinforcement Next Token Prediction with entropy and random token selection strategies.

- **Random Masked Span Reasoning.** A word span containing several tokens in the sequence is randomly selected and masked (Joshi et al., 2020), allowing the CoT to predict more than one tokens.
- **Entropy-based Next Token Reasoning.** The token with the top 20% entropy in the sequence is randomly selected and masked, with all subsequent tokens truncated, which is consistent with RPT (Dong et al., 2025).

Empirical Observation. Preliminary experiments are conducted to evaluate these masking strategies on the Wikipedia corpus, with performance measured by MMLU-Pro. According to Figure 9 in Appendix.A, Random NTP RLPT outperforms Mask RLPT with more stable training:

Finding 1. *Although Mask RLPT increases the predicted tokens, the vanilla random word-span selection strategy cannot effectively capture richer semantics in pretraining.*

To further investigate the effect of token selection, Random NTP is compared with Entropy NTP, where the token with higher entropy is selected for masking. As shown in Figure 3, Entropy NTP leads to training collapse and rapid reward degradation. At the position marked *stop optimization*, the reward signal becomes degenerate—all samples within a group yield either 0 or 1 accuracy. The reason is the data quality discrepancy between the synthetic and raw corpus. While entropy-based selection performs well on OmniMath (in RPT) (Gao et al., 2024), a high-quality synthetic dataset where high-entropy tokens consistently represent challenging but learnable patterns, the same strategy fails on Wikipedia. Raw Wikipedia data contains noise and inconsistencies, causing high-entropy tokens to be either genuinely difficult or simply noisy and unpredictable, which creates unstable learning signals:

Finding 2. *In real-world pretraining data distributions, selecting high-entropy tokens is no more effective than a random selection strategy, and learning actively from noisy*

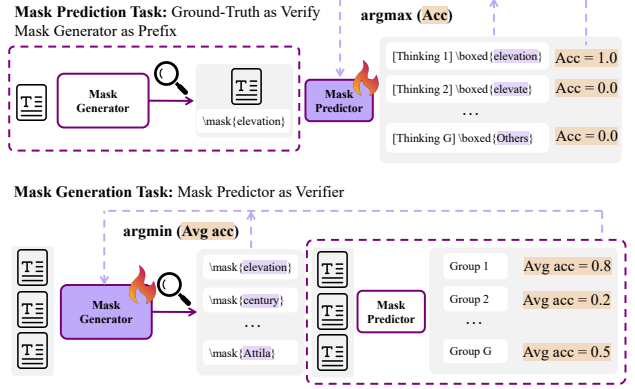


Figure 4. Pretraining Mask Prediction and Mask Generation tasks with GRPO (Shao et al., 2024).

data is necessary.

3.2. Active Pretraining Tasks

The limited performance of these passive masking strategies motivates a more informative and effective learning approach. Consider how humans learn: students focus on informative and valuable content in their experience that maximizes their improvement, rather than randomly selecting practice materials. Inspired by the active learning behavior, we propose an active masking strategy where the model learns to identify beneficial masking positions during training. Rather than relying on fixed heuristics like random sampling or entropy thresholding, the model discovers which tokens provide the strongest learning signals. The training process consists of two tasks for the shared LLM, as shown in Figure 4:

Mask Generation: given a text sequence s from pretraining data $\mathcal{D}$, the pretraining LLM π_ω first generates a thinking process and then generates a word span, $m \sim \pi_\omega(\cdot | s)$, to mask in this sequence. As shown in Figure 11 in Appendix.B, we initially prompt the policy $\pi_\omega(\cdot | s)$ to generate a span mask with one or several words verifiable for reasoning. During pretraining, the policy π_ω continuously learns to explore and capture semantic contents from the noisy pretraining corpus by RL. During the early training stage, the mask prediction policy is relatively weak and requires explicit clues, while in later stages, it needs to focus on the harder and unsolved words and domains.

Mask Prediction: We introduce the masked span prediction as a verifiable reinforcement learning task. Different from next token reasoning, a single CoT process predicts multiple masked tokens in a continuous span, $s_{[p:q]}$. Given the generated mask $m \sim \pi_\omega(\cdot | s)$, we replace the word span $s_{[p:q]}$ with the mark $[\text{mask}]$ in the sequence, and then recover the masked content through CoT reasoning. As shown in Figure 11, we prompt the policy $\psi_\omega(\cdot | m, s)$ to directly generate a CoT at the initial stage before the final

mask prediction $\hat{x} \sim \psi_\omega(\cdot | m, s)$. During optimization, the CoT reasons verifiable and semantic targets from the prefixed mask generation task.

3.3. Reinforcement Active Learning

Active learning objective. We cast mask generation and mask prediction as a coupled adversarial process, implemented with a shared LLM parameterized by ω . The generator $\pi_{\omega'}(\cdot | s)$ proposes masking patterns, while the predictor $\psi_\omega(\cdot | m, s)$ seeks to recover the masked content. Based on the final mask prediction rewards $R(s, m, \hat{x})$, this interaction is governed by the objective:

$$J(\omega) := \mathbb{E}_{\substack{s \sim \mathcal{D}, \\ m \sim \pi_{\omega'}(\cdot | s)}} \left[\mathbb{E}_{\hat{x} \sim \psi_\omega(\cdot | m, s)} [R(s, m, \hat{x})] \right], \quad (4)$$

which evaluates the predictor’s performance under the generator’s masking strategy. To encourage increasingly informative and challenging masks, we define the generator’s objective as $V(\omega) = \min_{\omega'} J(\omega)$, while the predictor optimizes in the opposite direction, i.e., $\arg \max_{\omega} J(\omega)$, thereby forming a coupled Min–Max formulation:

$$\begin{aligned} \omega^* &= \arg \max_{\omega} V(\omega) \\ &= \arg \min_{\omega' \in \Omega} \max_{\omega \in \Omega} \mathbb{E}_{s \sim \mathcal{D}} \left[\mathbb{E}_{\substack{m \sim \pi_{\omega'}(\cdot | s), \\ \hat{x} \sim \psi_\omega(\cdot | m, s)}} [R(s, m, \hat{x})] \right]. \end{aligned} \quad (5)$$

This adversarial min–max structure naturally mirrors the principle of active learning, where the generator actively selects reasonable and informative masks to probe the model’s weaknesses, thereby driving the predictor toward improved robustness and generalization.

Reinforcement optimization. To optimize the min–max active-learning objective in Eq. (5), we implement both mask prediction and mask generation as RL problems. For the prediction policy $\psi_\omega(\hat{x} | m, s)$, the reward is simply defined as an exact match between the predicted token span $\hat{x}^i$ and ground-truth x^i :

$$r_{\text{pred}}^i = R(s, m, \hat{x}) = \mathbb{I}[\hat{x}^i = x^i], \quad (6)$$

which directly optimizes inner maximization $\mathbb{E}_{\hat{x} \sim \psi_\omega(\cdot | m, s)} [R(s, m, \hat{x})]$ in Eq. (4). For the generation policy $\pi_{\omega'}(m | s)$, the reward is defined as the negative prediction accuracy under its own masks:

$$r_{\text{gen}}^j = 1 - \mathbb{E}_{x \sim \psi_\omega(\cdot | m, s)} [R(x)] = 1 - \frac{1}{G} \sum_{i=1}^G \mathbb{I}[\hat{x}^{i,j} = x^{i,j}], \quad (7)$$

which aligns with the outer minimization $\mathbb{E}_{\substack{s \sim \mathcal{D}, \\ m \sim \pi_{\omega'}(\cdot | s)}} \left[\mathbb{E}_{\hat{x} \sim \psi_\omega(\cdot | m, s)} [R(s, m, \hat{x})] \right]$ in Eq. (4). The mask generator is rewarded when its masks lead to a lower prediction accuracy, indicating that the induced masks

contain higher information content for the model. In addition, when the mask prediction accuracy is zero, we further define the generator’s reward to be $r_{\text{gen}}^j = 0$, to avoid rewarding noisy masks that are not predictable for $\psi_\omega(\cdot | m, s)$.

Given the reward definitions, r_{pred}^i and r_{gen}^j , we optimize Eq. (5) using GRPO. By substituting r_{gen}^j into the generator’s advantage, we obtain $A_{\text{gen}}^j = -\mathbb{E}[A_{\text{pred}}^{i,j}]$, which proves that the GRPO update is fully consistent with the Min–Max objective in Eq. (5):

$$\hat{A}_{\text{gen}}^j = \frac{r_{\text{gen}}^j - \text{mean}(r^1, \dots, r^G)}{\text{std}(r^1, \dots, r^G)} = -\mathbb{E}[\hat{A}_{\text{pred}}^{i,j}]. \quad (8)$$

We directly concatenate and uniformly optimize the mask generation and prediction batches in each step: $\mathcal{L}_{\text{GRPO}}(\omega) = -\frac{1}{G} \sum_{i=1}^G \min(r_i(\omega) \hat{A}_i, \text{clip}(r_i(\omega), 1 - \epsilon, 1 + \epsilon) \hat{A}_i)$.

4. Experimental Results

4.1. Implementation Details

Model. To evaluate stand-alone reinforcement pretraining, we directly continue pretraining the base models using reinforcement learning without introducing supervised fine-tuning (SFT) dataset. Specifically, we pretrain base models in 3 ~ 30 billion parameters, including the Qwen3-4B-Base, Qwen3-8B-Base, Qwen3-30B-A3B-MoE-Base, and SmolLM3-3B-Base (Bakouch et al., 2025), which is fully open-source, with all datasets and training recipes released.

Dataset. To evaluate on real-world distributed pretraining corpus, we use only the most general Wikipedia dataset. Existing RLPT often includes explicit QA pairs or synthetic datasets (e.g., OmniMath) with strong reasoning CoTs; this risks allowing the RL objective to copy these reasoning traces directly, implicitly degrading to supervised learning.

Training. For RLPT, we train 2000 steps using GRPO without KL regularization (Schulman et al., 2017). Following DAPO (Yu et al., 2025a), we filter samples whose accuracies are exactly 0.0 or 1.0, and adopt the clip-higher strategy for stability. For Qwen-Base models, we directly perform the PretrainZero strategy; for SmolLM3-3B-Base, we first use random RLPT for 100 steps as RL cold-start, and then perform PretrainZero for the remaining 1900 steps. During reasoning, the max length of the prompt and response is limited to 1536 and 4096 tokens, respectively. We adopt the 5×10^{-7} learning rate and the cosine scheme. In the mask-generation task, each batch contains 32 pretraining paragraphs, with 8 rollouts for each to produce masks. In the mask-prediction task, we evaluate 256 masks from the prefixed mask generation task (32×8), and each mask is also paired with 8 rollouts for prediction. Consequently, the overall prompt batch size becomes 288 for RL ($32 + 32 \times 8$).

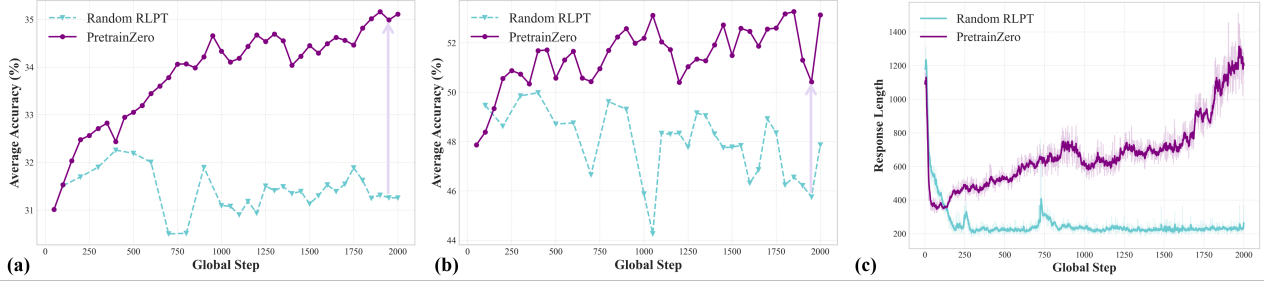


Figure 5. Comparisons between PretrainZero and Random RLPT on Qwen3-4B-Base: (a) the average accuracy on 3 general reasoning benchmarks; (b) the average accuracy on 6 math reasoning benchmarks; (c) response length of overall samples in training.

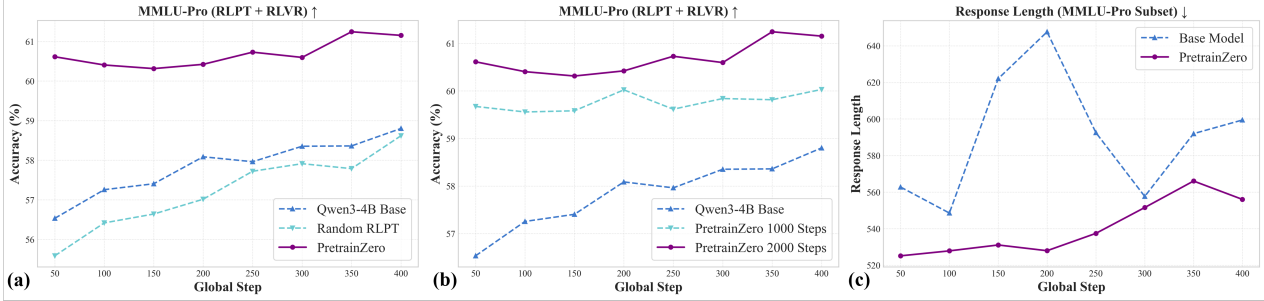


Figure 6. RLPT performance after the same RLVR post-training. (a) Comparison of Qwen3-4B-base, Random RLPT, and PretrainZero. (b) Comparison of Qwen3-4B-base, PretrainZero with 1000 and 2000 steps RLPT. (c) Response length comparison in MMLU-Pro subset.

Evaluation. We evaluate on both general and math reasoning benchmarks. For general domain, we evaluate on the MMLU-Pro (Wang et al., 2024a), SuperGPQA (Du et al., 2025), and BBEH (Kazemi et al., 2025); for math domain, we evaluate on 6 widely used benchmarks, including Math 500 (Hendrycks et al., 2021), Olympiad (He et al., 2024), Minerva (Lewkowycz et al., 2022), GSM8K (Cobbe et al., 2021), AMC23, and AIME24. For AMC23 and AIME24, we evaluate 32 times and report the mean@32 accuracy. We use the Qwen-Math-eval (Yang et al., 2024) as the verifier.

4.2. Pretraining Results

Baselines. To compare Reinforcement Learning on Pre-Training data (RLPT) with conventional training patterns, we primarily establish the following baselines: 1) the base model as the initial baseline. 2) Continued Pre-Training: We continuously pretrain with the self-supervised next token prediction on the same Wikipedia data. 3) Supervised Fine-Tuning: we formulate the masked token prediction task as question-answer pairs as Figure 11, and remove the CoT. 4) Random RLPT: We use the introduced random masked span prediction introduced in Sec. 3.1 as the strong RL baseline.

Comparison with Supervised Pretraining. We summarize general / math reasoning performance in Table 1. Compared with the base model, Continued Pre-Training and Supervised Fine-Tuning lead to overall performance drops of 16.47 and 8.09 on Qwen3-4B, and 24.75 and 10.27 on

Qwen3-8B, respectively. This occurs because, for highly optimized models, supervised learning on Wikipedia passages offers limited meaningful information and can even disrupt the model’s pretrained distribution. For RLPT, Random RLPT and PretrainZero improve 3.05 and 7.25 on Qwen3-4B, and 3.89 and 5.71 on Qwen3-8B. Comparisons suggest that RLPT is capable of extracting meaningful supervision from relatively low-quality data.

Comparison with Reinforcement Pretraining. As mentioned in Sec. 3.1, previous RPT training on high-entropy tokens quickly stops optimization on real-world pretraining corpus. We compare the training dynamics between Random RLPT and PretrainZero in Figure 5 and 10. As training steps increase, both the reasoning length of PretrainZero and its performance on general / math reasoning benchmarks improve consistently. This indicates that PretrainZero’s reasoning ability strengthens, similar to RLVR in DeepSeek-R1. Compared with Random RLPT, the active-learning strategy arouses longer CoT trajectories and noticeably stronger reasoning performance. PretrainZero consistently outperforms Random RLPT by 4.20, 1.82, 3.17, and 3.16 points on the Qwen3-4B, Qwen3-8B, Qwen3-30B-A3B-MoE, and SmolLM3-3B base models, respectively.

Reasoning Efficiency. As shown in Figure 5 (c), although the growth of CoT length in training, we need not worry about the inference efficiency of the reasoning process. The growth mainly comes from improvements in the mask-

Table 1. Results on math-domain (left) and general-domain (right) reasoning benchmarks. MATH indicates the average performance of the 6 math benchmarks. We highlight the best performance in **bold** and the second best in underline.

Method	Math-Domain Reasoning							General-Domain Reasoning				
	AVG	M500	Oly	Min	GSM	AMC	AIME	Overall	MATH	SGPQA	BBEH	MMLUP
<i>Qwen3-4B-Base</i>												
Base Model	42.53	73.30	37.30	22.10	86.30	36.17	0.00	32.36	42.53	26.32	8.67	51.94
Continue PT	24.65	38.00	13.60	11.00	67.00	15.00	3.30	15.89	24.65	9.67	0.04	29.21
Supervised FT	15.55	28.50	8.10	14.30	27.40	15.00	0.00	24.27	15.55	26.38	<u>12.28</u>	42.88
Random RLPT	<u>47.87</u>	<u>74.80</u>	<u>38.50</u>	<u>22.10</u>	<u>87.50</u>	<u>54.30</u>	<u>10.00</u>	<u>35.41</u>	<u>47.87</u>	<u>29.10</u>	9.45	<u>55.21</u>
PretrainZero	53.13	79.10	42.70	33.80	92.90	56.95	13.30	39.61	53.13	32.28	12.68	60.37
<i>Qwen3-8B-Base</i>												
Base Model	47.48	70.10	35.30	25.40	91.50	52.58	10.00	37.07	47.48	31.12	10.49	59.19
Continue PT	27.78	42.70	16.90	11.80	55.30	33.28	6.70	12.32	27.78	9.94	0.04	11.51
Supervised FT	19.23	30.50	11.70	15.40	32.80	25.00	0.00	26.80	19.23	29.02	11.17	47.78
Random RLPT	<u>55.08</u>	<u>79.20</u>	42.70	<u>39.00</u>	<u>93.80</u>	<u>62.50</u>	<u>13.30</u>	<u>40.96</u>	<u>55.08</u>	<u>34.19</u>	<u>12.96</u>	<u>61.59</u>
PretrainZero	57.72	81.90	<u>42.50</u>	43.40	93.50	65.00	20.00	42.78	57.72	34.46	14.67	64.28
<i>SmolLM3-3B-Base</i>												
Base Model	32.31	53.80	20.40	14.00	81.20	22.81	<u>1.65</u>	16.23	32.31	12.62	3.32	16.66
Random RLPT	<u>35.95</u>	<u>59.00</u>	<u>21.50</u>	<u>20.20</u>	<u>82.50</u>	32.50	0.00	<u>20.25</u>	<u>35.95</u>	<u>14.48</u>	7.85	<u>22.74</u>
PretrainZero	38.03	62.60	25.60	22.10	83.70	<u>27.50</u>	6.70	23.41	38.03	19.44	<u>3.78</u>	32.41
<i>Qwen3-30B-A3B-MoE-Base</i>												
Base Model	52.49	74.70	43.00	22.80	91.10	66.95	16.36	38.88	52.49	33.73	10.51	58.79
Random RLPT	<u>52.62</u>	<u>79.20</u>	41.20	<u>38.60</u>	82.40	59.77	14.58	<u>40.38</u>	<u>52.62</u>	<u>36.33</u>	<u>12.99</u>	<u>59.57</u>
PretrainZero	58.12	81.70	43.40	41.20	94.40	70.62	17.40	43.55	58.12	36.58	14.91	64.59

Table 2. Results on math-domain (left) and general-domain (right) reasoning benchmarks after the RLVR post-training. **RLPT / RLVR** indicates RL steps in RLPT and RLVR stages respectively. We highlight the best performance in **bold**.

Method	RLPT / RLVR	Math-Domain							General-Domain				
		AVG	M500	Oly	Min	GSM	AMC	AIME	Overall	MATH	SGPQA	BBEH	MMLUP
Base Model	- / 400	50.96	75.70	41.80	31.60	91.30	52.03	13.30	37.90	50.96	30.26	11.59	58.80
Random RLPT	2000 / 400	47.87	74.80	38.50	22.10	87.50	54.30	10.00	38.43	51.49	30.77	12.83	58.62
PretrainZero	1000 / 400	51.84	77.00	43.00	32.40	92.50	55.00	11.13	39.15	51.84	32.32	12.39	60.03
PretrainZero	2000 / 400	53.77	78.80	43.00	39.70	93.00	54.84	13.30	40.46	53.77	33.30	13.61	61.15

prediction capability. To verify this, we sample 10% of the MMLU-Pro prompts and evaluate the reasoning length. As shown in Figure 10 (c), for the same prompts, the reasoning length remains similar during RLPT. Moreover, compared with the base model, RLPT improves the CoT efficiency.

Reasoning Pattern. As shown in Figure 12, Appendix.B, we compare the reasoning patterns between Random RLPT and PretrainZero from the Qwen3-8B-Base. Given the same masked target, Random RLPT directly outputs the answer without any explicit reasoning. In contrast, PretrainZero first explores multiple possibilities, analyzes and verifies them step by step, and finally summarizes to reach a conclusion. The emergence of reasoning behavior during pretraining provides a stronger reasoning ability for generalization.

4.3. Post-Training Results

To investigate whether PretrainZero enhances the general reasoning capabilities of the foundation model for efficient RL finetuning, we apply RLVR as a post-training stage on PretrainZero. For general RLVR, we follow the General

Reasoner recipe (Ma et al., 2025), using the Web-Instruct dataset in a QA format and the same pretrained reward model as the verifier. For efficient RL finetuning, we train 400 steps on Qwen3-4B series models with a single node with 8× H800 GPUs, which supports at most the 128 batch-size, 1/8 compared with General Reasoner.

We evaluate the training process in Figure 6, and report the final general / math domain performance in Table 2. As shown in Figure 6 (b), the performance consistently improves as the training starts progressing from the base model to PretrainZero at 1000 RLPT steps and further to PretrainZero at 2000 RLPT steps on MMLU-Pro. As shown in Figure 6 (c), PretrainZero has more stable and efficient CoT in downstream RLVR. Compared with the base model, PretrainZero significantly improves the math average and overall accuracy by 2.18 and 2.56 points end-to-end.

4.4. Ablation Studies

Data Domain. To explore the impact of data domain on RLPT, we compare RLPT performance on the general-

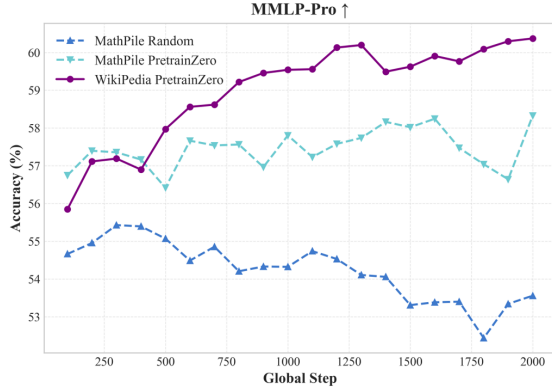


Figure 7. Comparisons for data domain. We evaluate MMLU-Pro performance on MathPile and Wikipedia.

domain Wikipedia and math-domain MathPile (Wang et al., 2024b). As shown in Figure 7, directly using general-domain Wikipedia data yields better performance, which result consists with General Reasoner (Ma et al., 2025). Since curating high-quality mathematical data requires substantial expert effort, we recommend using general-domain data for a much lower cost of data acquisition. To evaluate PretrainZero in post-training data, more results on OmniMath (Gao et al., 2024) shows that PretrainZero can still achieve better performance than RPT (Dong et al., 2025) (Appendix C).

Training Robustness. To confirm RLPT over 2000 steps, we evaluate different mask regularization strategies:

- 1) PretrainZero: For the generated masks, we retain only those whose underlying spans appear fewer than eight times within the paragraph.
- 2) PretrainZero-OneMask: Based on PretrainZero, if a generated mask appears multiple times in the paragraph, we randomly replace only one occurrence with [mask] and make prediction.
- 3) PretrainZero-Words: Since PretrainZero may produce masks that cover incomplete words—reducing interpretability—we filter masks that keep only complete word spans. As shown in Figure 8, three recipes can be trained stably, and PretrainZero consistently achieves better performance.

5. Related Works and Discussion

Self-Supervised Pretraining for LLMs. Scalable self-supervised pretraining (Kaplan et al., 2020) underlies modern large language models. With the fixed next-token prediction objective, LLMs (Vaswani et al., 2017; Team et al., 2025a) scale to Internet-level corpora and achieve strong general-purpose capabilities. Beyond this paradigm, masked-token objectives (Devlin et al., 2019; Joshi et al., 2019) remain important in BERT-style embeddings (Chen et al., 2024), diffusion language models (Nie et al., 2025), and code pretraining (Hui et al., 2024). The reliability, scalability, and broad applicability of self-supervised learning

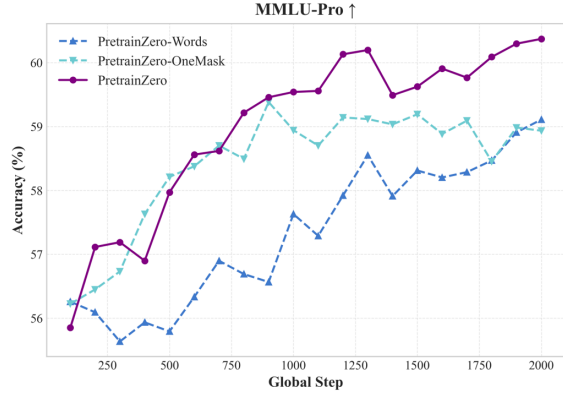


Figure 8. Comparisons for training robustness. We evaluate MMLU-Pro on different mask regularization strategies.

offer key insights for reinforcement pretraining and highlight its potential as a fundamental training strategy.

Reinforcement Learning for LLMs. Recent LLMs are largely driven by post-training RL, achieving human-expert performance in domains such as web agents (Wang et al., 2025; Team et al., 2025b), tool use (Patil et al.; Shang et al., 2025), software development (Jimenez et al., 2023), and mathematics (Shao et al., 2025). Despite this progress, existing RLHF (Ouyang et al., 2022) and RLVR (Lambert et al., 2025) rely on human annotation or domain-specific verification environments, causing a severe data bottleneck in general domains (Ma et al., 2025; Zhou et al., 2025; Yu et al., 2025b). RLHF requires continuously updated human-labeled data to prevent reward hacking, while RLVR is restricted to domains with verifiable ground-truth answers, fundamentally limiting its scalability for general reasoning.

Reinforcement Learning on Pretraining Data. To overcome the verification data wall, Reinforcement Learning on Pre-Training data (RLPT) has emerged as a promising direction, constructing general-purpose RLVR directly on pre-training corpora via self-supervised objectives. Early works such as Quiet-STaR (Zelikman et al., 2024) and Fast Quiet-STaR (Huang et al., 2025b) focus on token-level reasoning. Reinforcement Pre-Training (RPT) (Dong et al., 2025) is the first to adopt next-token prediction as an RLVR objective, demonstrating the feasibility of general-purpose RL. However, RPT remains limited by its reliance on synthetic OmniMath data with CoT annotations and post-trained initialization rather than base models, hindering RPT for practical and prolonged (Liu et al., 2025; He et al., 2025) RLPT.

Recently, PRT (Hatamizadeh et al., 2025) and RLPT¹ (Li et al., 2025) are proposed around the similar period as this work. PRT incorporates RL as an auxiliary objective to the self-supervised pretraining and does not exclude some QA-style training data. RLPT¹ introduces a reward-model verifier for segment-level prediction and a high-quality SFT cold start. Despite these advantages, the fundamental ques-

tions remain unexplored: under fully self-supervised conditions—removing reward models, SFT data, and supervised cross-entropy losses—can stand-alone RLPT be effectively trained on noisy, real-world pretraining corpus? And how to improve learning efficiency in low-information-density pretraining data? Addressing these fundamental questions becomes the primary focus of this work.

6. Conclusion

This work introduces the stand-alone reinforcement pretraining method for real-world pretraining corpus. Coupled with PretrainZero, a new reinforcement active Learning framework is proposed to explore informative, verifiable, and not-yet-mastered content in noisy pretraining data. Thanks to active learning ability, PretrainZero significantly surpasses previous fixed learning patterns, such as continued pretraining, supervised fine-tuning, and random or entropy-based reinforcement pretraining. We further show that even Wikipedia, already used in base model pretraining, can still improve end-task performance through reinforcement and active learning. We believe that there would be great potential to explore more efficient learning patterns to discover latent information from the pretraining corpus in the future.

Acknowledgments

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., Das-Sarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Bakouch, E., Ben Allal, L., Lozhkov, A., Tazi, N., Tunstall, L., Patiño, C. M., Beeching, E., Roucher, A., Reedi, A. J., Gallouédec, Q., Rasul, K., Habib, N., Fourier, C., Kydlicek, H., Penedo, G., Larcher, H., Morlon, M., Srivastav, V., Lochner, J., Nguyen, X.-S., Raffel, C., von Werra, L., and Wolf, T. SmolLM3: smol, multilingual, long-context reasoner. <https://huggingface.co/blog/smollm3>, 2025.
- Berglund, L., Tong, M., Kaufmann, M., Balesni, M., Stickland, A., Korbak, T., and Evans, O. The reversal curse: LLMs trained on “a is b” fail to learn “b is a”. *arXiv 2023. arXiv preprint arXiv:2309.12288*.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pp. 1877–1901. Curran Associates, Inc., 2020.
- Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., and Liu, Z. Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*, 2024.
- Chu, T., Zhai, Y., Yang, J., Tong, S., Xie, S., Schuurmans, D., Le, Q. V., Levine, S., and Ma, Y. Sft memorizes, rl generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL <https://www.aclweb.org/anthology/N19-1423>.
- Dong, Q., Dong, L., Tang, Y., Ye, T., Sun, Y., Sui, Z., and Wei, F. Reinforcement pre-training. *arXiv preprint arXiv:2506.08007*, 2025.
- Du, X., Yao, Y., Ma, K., Wang, B., Zheng, T., Zhu, K., Liu, M., Liang, Y., Jin, X., Wei, Z., et al. SuperGPQA: Scaling LLM evaluation across 285 graduate disciplines. *arXiv preprint arXiv:2502.14739*, 2025.
- Gao, B., Song, F., Yang, Z., Cai, Z., Miao, Y., Dong, Q., Li, L., Ma, C., Chen, L., Xu, R., Tang, Z., Wang,

- B., Zan, D., Quan, S., Zhang, G., Sha, L., Zhang, Y., Ren, X., Liu, T., and Chang, B. Omni-MATH: A universal Olympiad level mathematic benchmark for large language models. *ArXiv*, abs/2410.07985, 2024. URL <https://api.semanticscholar.org/CorpusID:273233571>.
- Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Hatamizadeh, A., Akter, S. N., Prabhumoye, S., Kautz, J., Patwary, M., Shoenybi, M., Catanzaro, B., and Choi, Y. Rlp: Reinforcement as a pretraining objective. *arXiv preprint arXiv:2510.01265*, 2025.
- He, B., Qu, Z., Liu, Z., Chen, Y., Zuo, Y., Qian, C., Zhang, K., Chen, W., Xiao, C., Cui, G., et al. Justrl: Scaling a 1.5 b llm with a simple rl recipe. *arXiv preprint arXiv:2512.16649*, 2025.
- He, C., Luo, R., Bai, Y., Hu, S., Thai, Z., Shen, J., Hu, J., Han, X., Huang, Y., Zhang, Y., et al. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3828–3850, 2024.
- Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D. X., and Steinhardt, J. Measuring mathematical problem solving with the math dataset. *ArXiv*, abs/2103.03874, 2021. URL <https://api.semanticscholar.org/CorpusID:232134851>.
- Huang, C., Yu, W., Wang, X., Zhang, H., Li, Z., Li, R., Huang, J., Mi, H., and Yu, D. R-zero: Self-evolving reasoning llm from zero data. *arXiv preprint arXiv:2508.05004*, 2025a.
- Huang, W., Xiong, Y., Ye, X., Deng, Z., Chen, H., Lin, Z., and Ding, G. Fast quiet-star: Thinking without thought tokens. *arXiv preprint arXiv:2505.17746*, 2025b.
- Hui, B., Yang, J., Cui, Z., Yang, J., Liu, D., Zhang, L., Liu, T., Zhang, J., Yu, B., Lu, K., et al. Qwen2. 5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024.
- Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., and Narasimhan, K. Swe-bench: Can language models resolve real-world github issues? *arXiv preprint arXiv:2310.06770*, 2023.
- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., and Levy, O. SpanBERT: Improving pre-training by representing and predicting spans. *arXiv preprint arXiv:1907.10529*, 2019.
- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., and Levy, O. Spanbert: Improving pre-training by representing and predicting spans. *Transactions of the association for computational linguistics*, 8:64–77, 2020.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models. *CoRR*, abs/2001.08361, 2020.
- Kazemi, M., Fatemi, B., Bansal, H., Palowitch, J., Anastasiou, C., Mehta, S. V., Jain, L. K., Aglietti, V., Jindal, D., Chen, Y. P., et al. Big-bench extra hard. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 26473–26501, 2025.
- Lambert, N., Morrison, J., Pyatkin, V., Huang, S., Ivison, H., Brahman, F., Miranda, L. J. V., Liu, A., Dziri, N., Lyu, S., Gu, Y., Malik, S., Graf, V., Hwang, J. D., Yang, J., Bras, R. L., Tafford, O., Wilhelm, C., Soldaini, L., Smith, N. A., Wang, Y., Dasigi, P., and Hajishirzi, H. Tulu 3: Pushing frontiers in open language model post-training, 2025. URL <https://arxiv.org/abs/2411.15124>.
- Lewkowycz, A., Andreassen, A., Dohan, D., Dyer, E., Michalewski, H., Ramasesh, V. V., Slone, A., Anil, C., Schlag, I., Gutman-Solo, T., Wu, Y., Neyshabur, B., Gur-Ari, G., and Misra, V. Solving quantitative reasoning problems with language models. *ArXiv*, abs/2206.14858, 2022. URL <https://api.semanticscholar.org/CorpusID:250144408>.
- Li, S., Li, K., Xu, Z., Huang, G., Yang, E., Li, K., Wu, H., Wu, J., Zheng, Z., Zhang, C., et al. Reinforcement learning on pre-training data. *arXiv preprint arXiv:2509.19249*, 2025.
- Liu, A., Feng, B., Xue, B., Wang, B., Wu, B., Lu, C., Zhao, C., Deng, C., Zhang, C., Ruan, C., et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- Liu, M., Diao, S., Lu, X., Hu, J., Dong, X., Choi, Y., Kautz, J., and Dong, Y. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. *arXiv preprint arXiv:2505.24864*, 2025.
- Ma, X., Liu, Q., Jiang, D., Zhang, G., Ma, Z., and Chen, W. General-reasoner: Advancing llm reasoning across all domains. *arXiv preprint arXiv:2505.14652*, 2025.
- Nie, S., Zhu, F., You, Z., Zhang, X., Ou, J., Hu, J., Zhou, J., Lin, Y., Wen, J.-R., and Li, C. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.
- OpenAI. Hello, gpt-4o. <https://openai.com/index/hello-gpt-4o>, 2024.

- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- Patil, S. G., Mao, H., Yan, F., Ji, C. C.-J., Suresh, V., Stoica, I., and Gonzalez, J. E. The berkeley function calling leaderboard (bfcl): From tool use to agentic evaluation of large language models. In *Forty-second International Conference on Machine Learning*.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language models are unsupervised multitask learners. 2019.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Settles, B. Active learning literature survey. 2009.
- Shang, N., Liu, Y., Zhu, Y., Zhang, L. L., Xu, W., Guan, X., Zhang, B., Dong, B., Zhou, X., Zhang, B., et al. rstar2-agent: Agentic reasoning technical report. *arXiv preprint arXiv:2508.20722*, 2025.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Shao, Z., Luo, Y., Lu, C., Ren, Z., Hu, J., Ye, T., Gou, Z., Ma, S., and Zhang, X. Deepseekmath-v2: Towards self-verifiable mathematical reasoning. *arXiv preprint arXiv:2511.22570*, 2025.
- Snell, C., Lee, J., Xu, K., and Kumar, A. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- Team, K., Bai, Y., Bao, Y., Chen, G., Chen, J., Chen, N., Chen, R., Chen, Y., Chen, Y., Chen, Y., et al. Kimi k2: Open agentic intelligence. *arXiv preprint arXiv:2507.20534*, 2025a.
- Team, T. D., Li, B., Zhang, B., Zhang, D., Huang, F., Li, G., Chen, G., Yin, H., Wu, J., Zhou, J., et al. Tongyi deepresearch technical report. *arXiv preprint arXiv:2510.24701*, 2025b.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, 4-9 December 2017, Long Beach, CA, USA*, pp. 6000–6010, 2017.
- Wang, W., Xu, X., An, W., Dai, F., Gao, W., He, Y., Huang, J., Ji, Q., Jin, H., Li, X., et al. Let it flow: Agentic crafting on rock and roll, building the rome model within an open agentic learning ecosystem. *arXiv preprint arXiv:2512.24873*, 2025.
- Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024a.
- Wang, Z., Li, X., Xia, R., and Liu, P. Mathpile: A billion-token-scale pretraining corpus for math. *Advances in Neural Information Processing Systems*, 37:25426–25468, 2024b.
- Yang, A., Zhang, B., Hui, B., Gao, B., Yu, B., Li, C., Liu, D., Tu, J., Zhou, J., Lin, J., et al. Qwen2. 5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.
- Yang, J., Lal, R. G., Bowden, J. C., Astudillo, R., Hameedi, M. A., Kaur, S., Hill, M., Yue, Y., and Arnold, F. H. Active learning-assisted directed evolution. *Nature Communications*, 16(1):714, 2025b.
- Yu, Q., Zhang, Z., Zhu, R., Yuan, Y., Zuo, X., Yue, Y., Fan, T., Liu, G., Liu, L., Liu, X., Lin, H., Lin, Z., Ma, B., Sheng, G., Tong, Y., Zhang, C., Zhang, M., Zhang, W., et al. DAPO: An open-source LLM reinforcement learning system at scale. *ArXiv*, abs/2503.14476, 2025a. URL <https://api.semanticscholar.org/CorpusID:277104124>.
- Yu, T., Ji, B., Wang, S., Yao, S., Wang, Z., Cui, G., Yuan, L., Ding, N., Yao, Y., Liu, Z., et al. Rlpr: Extrapolating rlvr to general domains without verifiers. *arXiv preprint arXiv:2506.18254*, 2025b.
- Yue, Y., Chen, Z., Lu, R., Zhao, A., Wang, Z., Yue, Y., Song, S., and Huang, G. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model?, 2025. URL <https://arxiv.org/abs/2504.13837>.
- Zelikman, E., Harik, G., Shao, Y., Jayasiri, V., Haber, N., and Goodman, N. D. Quiet-star: Language models can teach themselves to think before speaking. *arXiv preprint arXiv:2403.09629*, 2024.
- Zhou, X., Liu, Z., Sims, A., Wang, H., Pang, T., Li, C., Wang, L., Lin, M., and Du, C. Reinforcing general reasoning without verifiers. *arXiv preprint arXiv:2505.21493*, 2025.

A. More Comparison Results

Reinforcement Prediction Methods. To investigate whether MTP (Masked token Prediction) or NTP (Next Token Prediction) leads to more effective RLPT, we first compare the two token prediction strategies on MMLU-Pro. For MTP, each random mask corresponds to a complete word, which may consist of one or multiple tokens. The experimental results are shown in Figure 9.

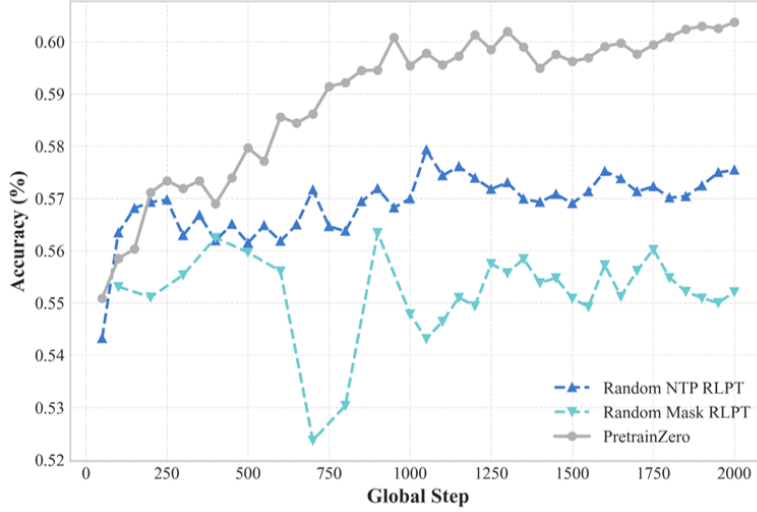


Figure 9. MMLU-Pro performance for reinforcement Random Next Token Prediction and Mask Token Prediction.

Training Dynamics in RLPT. We further compare the training dynamics of Random RLPT and PretrainZero, as shown in Figure 10. In terms of entropy, PretrainZero exhibits a larger exploration space than Random RLPT, which is more beneficial for generalization to downstream tasks. Regarding the overall difficulty of pretraining tasks, PretrainZero presents a more challenging on-policy learning setting than Random RLPT, showing a lower overall reward during training. We additionally evaluate reasoning length on the same validation set during training and observe that Random RLPT produces shorter reasoning trajectories than PretrainZero, accompanied by inferior performance. Finally, both PretrainZero and Random RLPT outperform the base model in terms of accuracy and CoT efficiency.

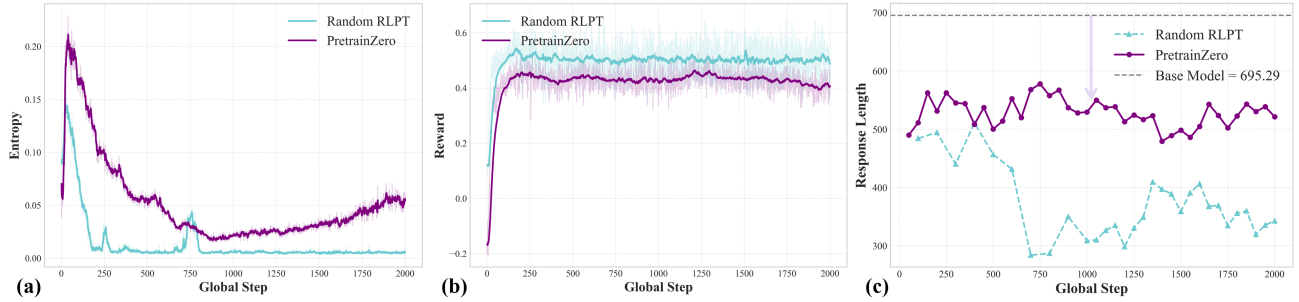


Figure 10. More comparisons between PretrainZero and Random RLPT on Qwen3-4B-Base: (a) entropy of model outputs; (b) the overall reward of pretraining tasks; (c) response length on a fixed subset from MMLU-Pro.

B. Task Templates and Reasoning Patterns

Prompt Example: Mask Generation / Prediction

Generate a mask to mask important words in the following paragraph, satisfying the requirements below:

- 1) The mask should mask one or more entities in the passage. The masked words should be continuous.
- 2) The masked words should exactly match words in the original passage.
- 3) The masked words could be predicted according to the context. The difficulty to predict should be moderately challenging for you, so the answer would be short and as unique as possible.

Paragraph: <paragraph>
The final generated masked words must be placed inside \mask{ }.

There is a passage with masked words by [mask]:
<paragraph with mask>
Please reason step by step, and put the predicted masked words within \boxed{ }.

Figure 11. Prompt for Mask Generation and Prediction.

Reasoning Patterns: Random RLPT / PretrainZero

Paragraph: In addition to his 1983 Triple Crown wins, Ralph Hanover won seventeen additional [mask] events, including the very important Adios and Meadowlands Pace ...

Random RLPT: Here is the passage with the masked words filled in:
In addition to his 1983 Triple Crown wins, Ralph Hanover won seventeen additional stakes ...
\boxed{stakes} ...

PretrainZero: To determine the masked word in the sentence:
"In addition to his 1983 Triple Crown wins, Ralph Hanover won seventeen ..."

1. **Analyze the sentence structure**: The sentence is: "Ralph Hanover won seventeen ..."
2. **Identify the missing word**: The sentence structure suggests that the missing word should complete the phrase "seventeen additional [mask] events." **Common phrases include** "seventeen additional stakes events" or "seventeen additional championship events."
3. **Consider the context**: The sentence mentions specific races like ... These are all racing events, particularly in harness racing. The mention of world records and specific track distances **further indicates that** the missing word is related to racing events.
4. **Determine the most appropriate word**: The most common completion ... in this context is ... **However, the most relevant term** in the context of racing achievements is "seventeen additional stakes events."
5. **Conclusion**: The missing word is "stakes."

Therefore, the predicted masked word is:
In addition to his 1983 Triple Crown wins, Ralph Hanover won seventeen additional \boxed{stakes} events ...

Figure 12. Reasoning pattern comparison between Random RLPT and PretrainZero.

C. Results on OmniMath

To further evaluate the effectiveness of PretrainZero under a data noise profile different from that of Wikipedia, we conduct 400 training steps on OmniMath (Gao et al., 2024) using DeepSeek-Distill-Qwen-7B as the base model. As shown in Table 3, although RPT (Dong et al., 2025) can be effectively optimized on OmniMath, PretrainZero achieves better average performance under the same training data and training steps, demonstrating its effectiveness across corpora with different noise profiles.

Method	Data	AMC23	GSM8K	MATH	SuperGPQA	BBEH	MMLU-Pro
DeepSeek-Distill-Qwen-7B	OmniMath	48.59	89.0	69.1	23.30	3.98	48.30
RPT	OmniMath	57.42	90.3	75.0	26.27	6.48	52.15
PretrainZero	OmniMath	56.17	90.8	80.6	27.81	8.34	54.24

Table 3. Comparison between PretrainZero and RPT on the OmniMath post-training data.